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# Adversarial Robustness and Functional Attacks in Neural Operators

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## Abstract

Neural Operators offer a robust solution to the computational challenges of Partial Differential Equations, delivering state-of-the-art efficiency in approximating infinite-dimensional mappings for physical systems. However, unlike traditional numerical solvers with established error bounds, these models exhibit failure modes when subjected to imperceptible input changes. This study evaluates the adversarial robustness of Fourier, Factorized Fourier and Convolutional Neural Operators against Physically-Informed Projected Gradient attacks. We show that physically constrained boundary perturbations can induce significant error rates while remaining difficult to detect, revealing that current neural operator architectures lack the inherent robustness required to replace traditional numerical solvers in sensitive applications.

## 1. Introduction

Neural operators such as the Fourier Neural Operator (FNO) (Li et al., 2021), Factorized Fourier Neural Operator (FFNO) (Tran et al., 2023), and Convolutional Neural Operator (CNO) (Raonić et al., 2023) have emerged as powerful tools for solving partial differential equations (PDEs). They are major improvements on top of previous state of the art PDE solvers such as Physics-Informed Neural Networks (PINNs) (Cai et al., 2021) or more general purpose U-Net/ResNet (Li et al., 2021), offering resolution-invariant predictions and significantly faster inference than traditional numerical solvers. However, little attention has been given to robustness of these architectures. Adversarial perturbations, small input changes in the coefficient field or boundary conditions, can lead to disproportionate prediction errors. This is concerning for the reliability of neural operators in applications such as fluid dynamics and climate modeling (Li et al., 2021).

While adversarial robustness is well-studied for image classifiers, neural operators present unique challenges due to functional inputs and physical constraints, raising the question: How robust are these models to small input perturbations?

This work makes the following contributions:

1. **Systematic robustness comparison:** We evaluate the adversarial robustness of three neural operator architectures (FNO, FFNO, CNO) across two canonical PDEs (2D Darcy Flow and 2D Navier-Stokes), revealing architecture-dependent and problem-dependent vulnerabilities.
2. **Physics-informed stealth attacks:** We introduce a novel attack framework that augments standard PGD with physics loss (PDE residual) and boundary condition loss, generating adversarial perturbations that respect physical constraints and are harder to detect than standard attacks.
3. **Key empirical findings and practical implications:** We determined that it is possible to use Projected Gradient Descent to identify input perturbations that have an increased MSE error across all the models FNO, FFNO and CNO, the error rate from 2-3.5% to 9-16% depending on the  $L_2$  norm of the input perturbation. The models are especially vulnerable to perturbations on the boundary conditions. MVMO and Physically-Informed attacks have a less impact on the MSE error while also requiring a smaller perturbation magnitude.

## 2. Models and Methods

### 2.1. Related Work

Adversarial vulnerability is well-established in image classification tasks (Szegedy et al., 2014; Goodfellow et al., 2015; Madry et al., 2017), yet remains under-explored for Neural Operators such as FNO (Li et al., 2021), FFNO (Tran et al., 2023), and CNO (Raonić et al., 2023) which map between infinite-dimensional function spaces. While (Kovachki et al., 2021) discuss theoretical error bounds for FNOs, empirical studies indicate significant vulnerabilities. (Adesoji & Chen, 2022) demonstrated that PGD attacks effectively degrade FNO performance on fluid dynamics tasks. Similarly, (Goswami et al., 2022) showed that Physics-Informed Neural Networks (PINNs) remain susceptible to attacks despite training with physics constraints. (Zhao et al., 2023) formalized functional adversarial attacks using Sobolev norms but did not incorporate physical constraints into the attack generation itself.

While recent research uses PGD primarily for robust training (Sun, 2025), we focus on characterizing the attacks themselves across both spectral and spatial domains. We advance this line of inquiry by: (1) systematically benchmarking FNO, FFNO, and CNO robustness on Darcy Flow and Navier-Stokes; and (2) introducing physics-informed "stealth" attacks that enforce boundary and PDE residual constraints to generate physically plausible perturbations.

## 2.2. Objectives and methodology

**Experimental Setup.** We evaluate robustness on synthetic datasets generated via Gaussian Random Fields (GRF) for two canonical systems: the steady-state *Darcy Flow* and the time-dependent *Navier-Stokes* equations. We trained FNO, FFNO, and CNO models for both problems using a composite loss function comprising data MSE and physics constraints (PINO (Li et al., 2023)).

To ensure rigorous baselines, all models were trained to convergence with low validation errors (approx. 2–3.5% for Darcy and < 3% for Navier-Stokes). We utilized 1000 training samples for Darcy Flow and 10,000 for Navier-Stokes. Detailed dataset generation parameters, numerical solver specifications, optimizer settings, and exact model hyperparameters are provided in Appendix B.

**Attack Methodology.** We evaluate the robustness of each model under the following scenarios:

*Random noise baselines.* We add random spatial and spectral noise with varying  $L_2$  perturbation magnitudes to establish baseline sensitivity.

*Pure PGD attacks.* We perform standard PGD attacks with various  $\epsilon$  parameters, iteratively maximizing output error without physical constraints.

*Physics-informed stealth attacks.* We augment the PGD loss with physics residual (PDE loss) and boundary condition loss to generate adversarial perturbations that respect physical constraints.

*MVMO (Raff et al., 2025), only for the Darcy Flow* We alternate between the PGD loss and the physics residual loss depending on whether the physics residual loss increases beyond a specified threshold.

*Boundary condition attacks, only for the Darcy Flow.* A flavor of PGD limited only to the perturbations on the boundaries of the input.

For Navier-Stokes, we additionally evaluate temporal attack propagation: since well-trained models can naturally resolve initial perturbations over time Figure 1, we test scenarios where perturbations are injected at every autoregressive timestep to observe error accumulation.

We document the perturbations and errors in their spatial

and spectral representations to understand whether these perturbations are simply observed as high-frequency noise, and evaluate the cross-resolution transferability of the attacks.

**Evaluation Metrics.** We evaluate attack effectiveness using:

- **Relative  $L_2$  Error (%)**:  $\frac{\|u_{\text{pred}} - u_{\text{true}}\|_2}{\|u_{\text{true}}\|_2} \times 100\%$
- **Amplification Factor**:  $\frac{\text{Output Error (\%)}}{\text{Input Perturbation (\%)}}$

## 2.3. Physically Informed Projected Gradient Descent

Given a model  $u$  and a sample  $x$ , standard PGD finds a perturbation  $\beta$  with  $\|\beta\|_\infty \leq \epsilon$  that maximizes

$$\|u(x) - u(x + \beta)\|_2^2. \quad (1)$$

If the attack is successful, the prediction of the adversarial sample  $\tilde{x} = x + \beta$  has a high error. However, these perturbations often force the model to violate fundamental physical constraints and lead to atypical predictions  $u(\tilde{x})$ . For example, in the 2D Darcy Flow problem, any solution  $u(x)$  must fulfill the boundary conditions. The adversarial prediction  $u(\tilde{x})$  may violate this condition and if it does it can easily be identified as adversarial. We therefore introduce the physics-informed loss

$$\begin{aligned} \mathcal{L}_{PI}(\beta) = & \text{MSE}(u(x), u(x + \beta)) \\ & - \lambda_{PDE} \mathcal{L}_{PDE}(u(x + \beta)) \\ & - \lambda_{BC} \mathcal{L}_{BC}(u(x + \beta)). \end{aligned} \quad (2)$$

$\mathcal{L}_{PDE}$  measures how well a prediction matches the underlying PDE and punishes predictions that obviously violate the PDE.  $\mathcal{L}_{BC}$  punishes predictions that violate the boundary conditions.  $\lambda_{PDE}$  and  $\lambda_{BC}$  are hyperparameters. Maximizing the stealth loss  $\mathcal{L}_{PI}$  leads to adversarial samples and predictions that are difficult to identify as such while retaining a large prediction error.

For the Darcy Flow equation we used

$$\mathcal{L}_{BC}(u(x)) = \frac{1}{|\partial D|} \sum_{x \in \partial D} u(x)^2. \quad (3)$$

$$\mathcal{L}_{PDE} = \frac{1}{N} \sum_{i=1}^N \left\| \underbrace{-\nabla \cdot (a \nabla u)}_{\text{Spectral Diff.}} - \underbrace{f}_{\text{Forcing constant}} \right\|_2^2 \quad (4)$$

For the Navier Stokes equation, we did not use a boundary condition loss as our instance of the problem does not have a boundary condition. As PDE-loss we used

$$\mathcal{L}_{PDE} = \frac{1}{N} \sum_{i=1}^N \left\| \underbrace{\frac{w_t - w_{t-1}}{\Delta t}}_{\text{Time Diff.}} + \underbrace{(\mathbf{u} \cdot \nabla) w_t}_{\text{Advection}} - \underbrace{\nu \Delta w_t}_{\text{Diffusion}} \right\|_2^2. \quad (5)$$

## 2.4. Spectral and spatial perturbations

The adversarial perturbation  $\beta$  can be selected in different ways. Given a sample  $x \in \mathbb{R}^d$ , the perturbation can be selected from the spatial domain  $\mathbb{R}^d$ . The adversarial sample then is  $\tilde{x} = x + \beta$ . This approach gives rise to spatial PGD attacks.

An alternative approach is to select the perturbation in the frequency domain. In this case the adversarial sample is given by  $\tilde{x} = \mathcal{F}^{-1}(\mathcal{F}(x) + \beta)$ .

This approach seems natural since FNOs make heavy use of Fourier transforms and operate mostly in the frequency domain. In our particular experimental setup, we intentionally higher frequencies target ( $6 \leq \|\mathbf{k}\|_2 \leq 12$ , where  $\mathbf{k}$  is the wavenumber).

## 3. Results

### 3.1. Darcy Flow

Table 1 (detailed in Appendix Table 7) reveals that pure PGD attacks yield the highest error amplification ( $\sim 2.5\times$ ), with spectral perturbations proving more effective than spatial ones by exploiting high frequencies within the  $\epsilon$ -bound. Vulnerability is consistent across architectures, with FNO ( $2.5\times$ ), FFNO ( $2.0\times$ ), and CNO ( $3.0\times$ ) showing similar fragility. As shown in Table 2, while MVMO aligns with Stealth performance, Boundary attacks demonstrate the highest amplification factors. Although absolute error magnitudes remain low ( $\sim 2.5\text{--}6\%$ ), this extreme sensitivity highlights a critical blind spot despite the inclusion of boundary condition losses during training.

Table 1. Spatial vs Spectral Attacks on Darcy Flow

| Model | Method  | Spatial |      |            | Spectral |     |            |
|-------|---------|---------|------|------------|----------|-----|------------|
|       |         | In      | Out  | Amp        | In       | Out | Amp        |
| CNO   | Clean   | 0       | 2.0  | 1.0        | 0        | 2.0 | 1.0        |
|       | Noise   | 5.2     | 2.6  | 0.5        | 5.1      | 2.5 | 0.5        |
|       | PGD     | 4.5     | 16.4 | <b>3.6</b> | 1.0      | 3.0 | <b>2.9</b> |
|       | Stealth | 2.7     | 3.5  | 1.3        | 1.0      | 2.4 | 2.3        |
| FNO   | Clean   | 0       | 3.5  | 1.0        | 0        | 3.5 | 1.0        |
|       | Noise   | 5.2     | 5.6  | 1.1        | 5.1      | 5.6 | 1.1        |
|       | PGD     | 4.6     | 11.5 | <b>2.5</b> | 1.0      | 5.9 | <b>5.8</b> |
|       | Stealth | 3.2     | 5.5  | 1.7        | 1.0      | 5.6 | 5.4        |
| FFNO  | Clean   | 0       | 2.9  | 1.0        | 0        | 2.9 | 1.0        |
|       | Noise   | 5.2     | 3.9  | 0.8        | 5.1      | 3.9 | 0.8        |
|       | PGD     | 4.6     | 9.2  | <b>2.0</b> | 1.0      | 4.1 | <b>4.0</b> |
|       | Stealth | 2.5     | 3.9  | 1.5        | 1.0      | 3.9 | 3.8        |

### 3.2. Navier Stokes

Table 3 summarizes error amplification results (full parameters in Appendix Table 8). The same pattern holds for Navier-Stokes, where PGD attacks consistently outperform random noise. Generally, spatial attacks have the highest er-

Table 2. Darcy Flow: Boundary & MVMO Attacks

| Model | Method   | In  | Out | Amp        |
|-------|----------|-----|-----|------------|
| CNO   | Clean    | 0.0 | 2.0 | 1.0        |
|       | Boundary | 0.8 | 2.6 | <b>3.3</b> |
|       | MVMO     | 2.7 | 3.5 | 1.3        |
| FNO   | Clean    | 0.0 | 3.5 | 1.0        |
|       | Boundary | 0.8 | 5.9 | <b>7.4</b> |
|       | MVMO     | 3.2 | 5.5 | 1.7        |
| FFNO  | Clean    | 0.0 | 2.9 | 1.0        |
|       | Boundary | 0.8 | 4.3 | <b>5.3</b> |
|       | MVMO     | 2.5 | 3.8 | 1.5        |

Table 3. Spatial vs Spectral Attacks on Navier-Stokes

| Model | Method  | Spatial |      |            | Spectral |     |            |
|-------|---------|---------|------|------------|----------|-----|------------|
|       |         | In      | Out  | Amp        | In       | Out | Amp        |
| CNO   | Clean   | 0       | 3.0  | 1.0        | 0        | 3.0 | 1.0        |
|       | Noise   | 5.2     | 2.7  | 0.5        | 5.2      | 2.7 | 0.5        |
|       | PGD     | 4.5     | 7.2  | <b>1.6</b> | 0.7      | 2.6 | <b>3.8</b> |
|       | Stealth | 7.0     | 5.8  | 0.8        | 1.4      | 2.2 | 1.5        |
|       | Seq.    | 0.0     | 15.7 | -          | 0.0      | 2.8 | -          |
| FNO   | Clean   | 0       | 0.7  | 1.0        | 0        | 0.7 | 1.0        |
|       | Noise   | 5.2     | 0.9  | 0.2        | 5.2      | 0.8 | 0.1        |
|       | PGD     | 9.1     | 9.2  | <b>1.0</b> | 0.6      | 0.7 | <b>1.2</b> |
|       | Stealth | 8.6     | 7.9  | 0.9        | 1.2      | 0.8 | 0.6        |
|       | Seq.    | 0.0     | 16.0 | -          | 0.0      | 1.7 | -          |
| FFNO  | Clean   | 0       | 0.6  | 1.0        | 0        | 0.6 | 1.0        |
|       | Noise   | 5.2     | 0.7  | 0.1        | 5.2      | 0.6 | 0.1        |
|       | PGD     | 9.0     | 9.0  | <b>1.0</b> | 0.6      | 0.5 | <b>0.9</b> |
|       | Stealth | 8.1     | 7.4  | 0.9        | 1.2      | 0.6 | 0.5        |
|       | Seq.    | 0.0     | 15.7 | -          | 0.0      | 1.2 | -          |

ror magnitude, while the spectral ones appear most effective with respect to the amplification ratio ( $1.2\times\text{--}3.8\times$ ). The spectral attacks have a significantly smaller error amplification attack regardless of the PGD  $\epsilon$ . Physically-Informed stealth attacks were less successful (see Table 4) on Navier-Stokes, however the PDE does not have similarly exploitable boundaries as the Darcy flow.

Temporal analysis (Figure 1) reveals that PGD attacks are primarily effective on short time scales. FFNO and FNO exhibit resilience, gradually reducing adversarial error to near-baseline levels over time. CNO is vulnerable only momentarily; over longer time scales, the clean sample has an even higher error than the adversarial ones.

### 3.3. Spectral aspects of the perturbation

Figure 2 shows the low/high frequency split of the input perturbations and output errors of the attacks showing the correlation between high frequency inputs to low frequency output. We expected this result, as we believed that the perturbations generally manifest themselves as high frequency noise on the inputs, but do cause significant low-frequency errors.

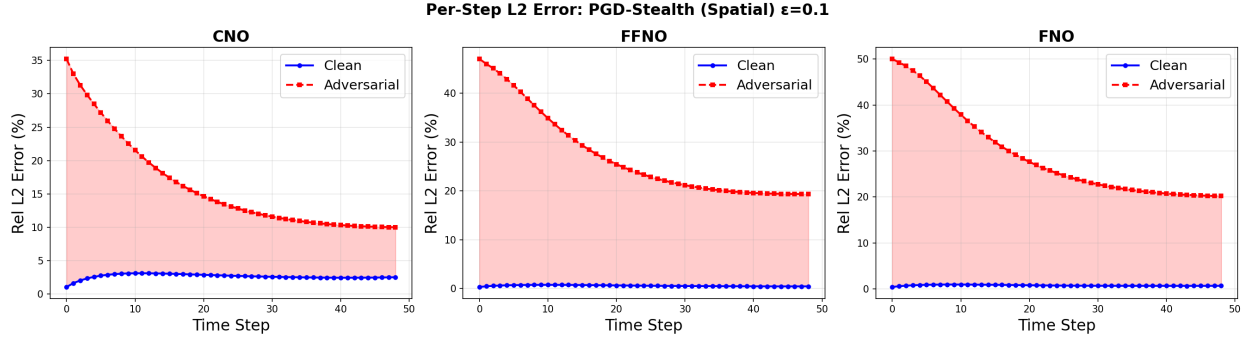
Figure 1. Plot of the relative  $L_2$  error for every time step of different autoregressive models.

Table 4. PGD-Pure vs PGD-Stealth on Navier-Stokes

| Model | Dom.  | Pure |     | Stealth |     |
|-------|-------|------|-----|---------|-----|
|       |       | In   | Out | In      | Out |
| CNO   | Spat. | 4.5  | 7.2 | 7.0     | 5.8 |
|       | Spec. | 0.7  | 2.6 | 1.4     | 2.2 |
| FNO   | Spat. | 9.1  | 9.2 | 8.6     | 7.9 |
|       | Spec. | 0.6  | 0.7 | 1.2     | 0.8 |
| FFNO  | Spat. | 9.0  | 9.0 | 8.1     | 7.4 |
|       | Spec. | 0.6  | 0.5 | 1.2     | 0.6 |

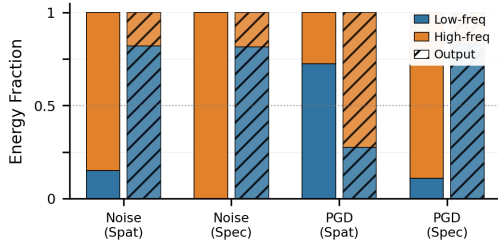


Figure 2. Spectral comparison of the input perturbations vs the output errors.

### 3.4. Cross-resolution transferability of the attacks

All attacks transfer positively — adversarial perturbations crafted at  $128 \times 128$  become more effective when evaluated at different resolutions. We conducted the attack at a  $128 \times 128$  Resolution, and evaluated on downsampled and upsampled instances of the same input conditions. Table 5 shows a comparison of the PGD spatial and spectral attacks across resolutions. Both the Darcy Flow and Navier-Stokes have a similar Cross-Resolution transferability properties. We can observe the following: Spectral attacks are more resolution-agnostic — they transfer 50% better, downsampling amplifies attacks more — lower-resolution models are more vulnerable to transferred attacks and CNO is anomalously vulnerable — spectral attacks show  $3.2 \times$  transfer to low res (vs  $1.5 \times$  for FNO).

Table 5. Cross-Resolution Transferability of Adversarial Attacks on Darcy Flow.

| Model | Domain   | Error (%) |      |      | Transfer                        |                                 |
|-------|----------|-----------|------|------|---------------------------------|---------------------------------|
|       |          | 128       | 64   | 256  | 128→64                          | 128→256                         |
| CNO   | Spatial  | 16.4      | 20.0 | 18.3 | $1.22 \times$                   | $1.12 \times$                   |
|       | Spectral | 3.0       | 9.7  | 6.3  | <b><math>3.20 \times</math></b> | <b><math>2.06 \times</math></b> |
| FNO   | Spatial  | 11.5      | 12.2 | 14.5 | $1.06 \times$                   | $1.26 \times$                   |
|       | Spectral | 5.9       | 9.4  | 9.0  | $1.58 \times$                   | $1.52 \times$                   |
| FFNO  | Spatial  | 9.2       | 10.6 | 12.6 | $1.15 \times$                   | $1.37 \times$                   |
|       | Spectral | 4.1       | 8.8  | 7.7  | $2.15 \times$                   | $1.89 \times$                   |

## 4. Discussion

Our results indicate that Neural Operators, despite their computational efficiency, currently lack the robustness required for safety-critical applications, exhibiting error amplification up to  $3.6 \times$  under standard PGD. Crucially, we observe a clear trade-off between attack stealth and effectiveness: perturbations constrained to respect physical laws (Stealth PGD) cause significantly less damage than unconstrained attacks. This suggests that the most destructive adversarial modes lie off the physical manifold, implying that simple physics-based residual checks could serve as a viable defense strategy. However, the extreme sensitivity to boundary perturbations in the elliptic Darcy Flow highlights a specific architectural weakness: models appear to overfit to boundary distributions rather than robustly learning the global propagation of boundary conditions.

Our analysis was limited by computational resources to fixed attack hyperparameters and 2D regimes. Future work should extend this to 3D turbulence and adaptive attack strategies to determine if “stealth” attacks can be optimized further, or if adversarial training can effectively close these robustness gaps.

All experiment data is available on [Polybox](#), and the source code can be found on [GitHub](#).

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## A. Appendix: Mathematical background

Neural Operators aim to learn a mapping between two infinite-dimensional function spaces, the input space  $\mathcal{A} = \mathcal{A}(D; \mathbb{R}^{d_a})$  and the output space  $\mathcal{U} = \mathcal{U}(D; \mathbb{R}^{d_u})$  - a solution to a differential equation (PDE) of the form:

$$(\mathcal{L}_a u)(x) = f(x), \quad x \in D \quad (6)$$

We define the solution operator  $G^\dagger : \mathcal{A} \rightarrow \mathcal{U}$  as the map taking a discretized parameter  $a$  to the discretized solution  $u$ . Our goal is to approximate  $G^\dagger$  with a neural operator  $G_\theta$  parameterized by  $\theta$ , such that:

$$G_\theta(a) \approx u \quad (7)$$

The optimization objective is to minimize the empirical risk over a dataset of  $N$  pairs  $\{(a_j, u_j)\}_{j=1}^N$ :

$$\min_{\theta} \frac{1}{N} \sum_{j=1}^N \|G_\theta(a_j) - u_j\|_2 \quad (8)$$

### A.1. Fourier Neural Operator

Given a particular input function

$$v_{l+1}(x) = \sigma(W v_l(x) + (\mathcal{K} v_l)(x)) \quad (9)$$

where  $\sigma$  is a non-linear activation function,  $W$  is a linear transformation (point-wise convolution), and  $\mathcal{K}$  is the kernel integral operator. In FNO,  $\mathcal{K}$  is parameterized in the frequency domain using the Fast Fourier Transform (FFT):

$$(\mathcal{K} v_l)(x) = \mathcal{F}^{-1}(R_\phi \cdot \mathcal{F}(v_l))(x) \quad (10)$$

### A.2. Factorized Fourier Neural Operators

While FNO parameterizes the spectral weight tensor  $R_\phi$  fully, the Factorized FNO (FFNO) imposes a low-rank structure on these weights to reduce computational complexity and overfitting. This is typically achieved via a tensor decomposition (e.g., Tucker decomposition) or by separating the mixing of Fourier modes and channels. The spectral operation becomes:

$$(\mathcal{K}_{\text{fact}} v_l)(x) = \mathcal{F}^{-1}((U \otimes V \otimes \dots) \cdot \mathcal{F}(v_l))(x) \quad (11)$$

where the full tensor  $R_\phi$  is approximated by the contraction of smaller tensors (e.g.,  $U, V$ ). This allows the FFNO to significantly reducing the number of parameters, enabling the training of deeper architectures on high-resolution data.

### A.3. Convolutional Neural Operators

The CNO is a resolution-independent architecture designed to address aliasing errors common in standard CNNs when applied to PDE solving. CNO relies on integral kernels. The operator maps a function  $v_l$  to  $v_{l+1}$  via:

$$v_{l+1}(x) = \sigma \left( \int_{B(x,r)} \kappa_\theta(x-y) v_l(y) dy \right) \quad (12)$$

where the kernel  $\kappa_\theta$  has finite support radius  $r$ . To ensure resolution invariance and discrete-continuous consistency, CNO implements the integral via discrete convolutions combined with signal reconstruction so that:

$$\text{CNO}(P_N u) \approx P_N(\text{CNO}(u)) \quad (13)$$

where  $P_N$  is a projection onto a discretization grid, which prevents high-frequency aliasing errors from propagating through the network layers.



#### A.4. 2D Darcy and 2D Navier Stokes Models

The steady-state Darcy Flow equation on the unit box domain  $D = [0, 1]^2$ . The system is governed by the second-order partial differential equation:

$$-\nabla \cdot (a(x) \nabla u(x)) = f(x), \quad x \in D \quad (14)$$

subject to boundary conditions  $u(x) = 0$  for  $x \in \partial D$ .

The goal is to learn the operator  $G^\dagger : a \mapsto u$ .

The 2D Navier-Stokes equation for a viscous, incompressible fluid in vorticity form on the domain  $D = [0, 1]^2$  (periodic boundary conditions), where the vorticity function  $w(x, t)$  is given by:

$$\begin{aligned} \frac{\partial w}{\partial t} + (v \cdot \nabla)w &= \nu \Delta w + f, \quad x \in D, t \in (0, T] \\ \nabla \cdot v &= 0 \end{aligned} \quad (15)$$

where  $v(x, t)$  is the velocity field,  $w = \nabla \times v$  is the vorticity,  $\nu$  is the viscosity parameter, and  $f$  is the forcing term.

The goal is to learn the solution operator that maps the initial state to the future state:  $G^\dagger : w(\cdot, t) \mapsto w(\cdot, t + \Delta t)$ , for an arbitrary temporal resolution defined by  $\Delta t$ .

## B. Appendix: Experimental Details and Hyperparameters

**Data Generation.** For the steady-state *Darcy Flow*, we map piecewise-constant permeability fields  $a(x) \in \{1, 12\}$  (derived from binarized GRFs) to pressure solutions  $u(x)$  under a constant forcing term  $f = 1$ .

For the time-dependent *Navier-Stokes* equations, we simulate vorticity  $w(x, t)$  on a torus with viscosity  $\nu = 10^{-3}$  from GRF initial conditions. The system is evolved up to  $t = 50$  using a pseudo-spectral Crank-Nicolson method with deterministic forcing to maintain flow energy:

$$f(x, y) = 0.1(\sin(2\pi(x + y)) + \cos(2\pi(x + y))) \quad (16)$$

**Training Configuration.** All models were trained with physics loss ( $\lambda_{BC} = \lambda_{PDE}$ ) in addition to MSE loss. We used the AdamW optimizer with weight decay  $1e^{-6}$  and a step learning rate scheduler with step size 100 and  $\gamma = 0.1$ . The specific model architectures are detailed in Table 6.

Table 6. Model hyperparameters and training configuration. Note that all models use 4 layers and a learning rate of  $1 \times 10^{-3}$ .

| MODEL                | MODES | WIDTH | IN SIZE | $\lambda_{PDE}$ | EPOCHS |
|----------------------|-------|-------|---------|-----------------|--------|
| <b>DARCY FLOW</b>    |       |       |         |                 |        |
| FNO                  | 32    | 64    | –       | $10^{-4}$       | 500    |
| FFNO                 | 32    | 64    | –       | $10^{-4}$       | 500    |
| CNO                  | –     | –     | 128     | $10^{-5}$       | 1000   |
| <b>NAVIER-STOKES</b> |       |       |         |                 |        |
| FNO                  | 12    | 20    | –       | 0.1             | 500    |
| FFNO                 | 16    | 64    | –       | 0.1             | 500    |
| CNO                  | –     | –     | 64      | 0.1             | 500    |

All models were trained with physics loss ( $\lambda_{BC} = \lambda_{PDE}$ ) in addition to MSE loss. We used the AdamW optimizer with weight decay  $1e^{-6}$  and a step learning rate scheduler with step size 100 and  $\gamma = 0.1$ . The specific model architectures are detailed in Table 6.

**Clean baseline.** Figure 5 Shows the training progress, with the final validation errors for FNO, FFNO and CNO being 3.5%, 2.9% and 2.0% respectively on the Darcy Flow and 0.7%, 0.58% and 2.99% on Navier Stokes.

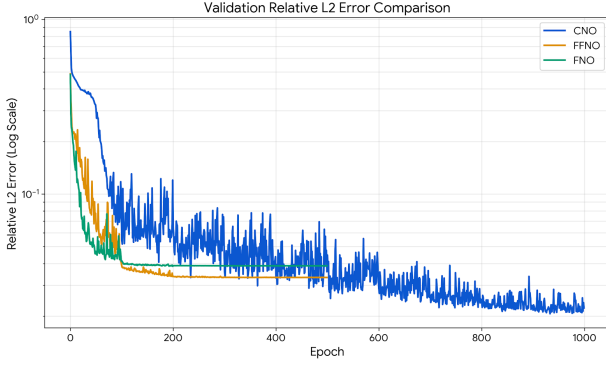


Figure 3. Darcy Training

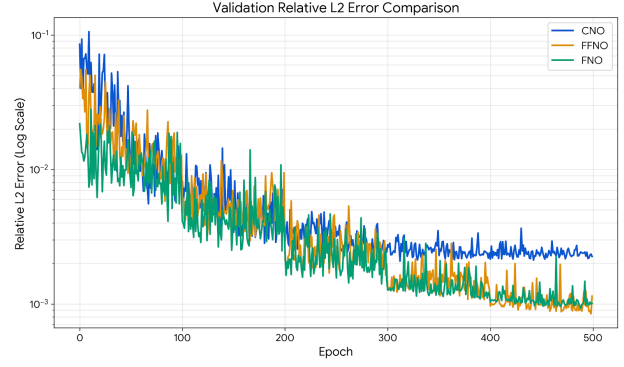


Figure 4. Navier-Stokes

Figure 5. Training curves.

### C. Appendix: Comprehensive Darcy Flow and Navier Stokes Results

**Darcy Flow.** Figure 6 below shows a comparison of the top attacks by efficiency (amplification factor), based on the relative values of the  $L_2$  perturbation magnitudes. Note that the values in the report are averaged across the attack types, while here, the variance of the efficacy within a single attack type can be more obvious.

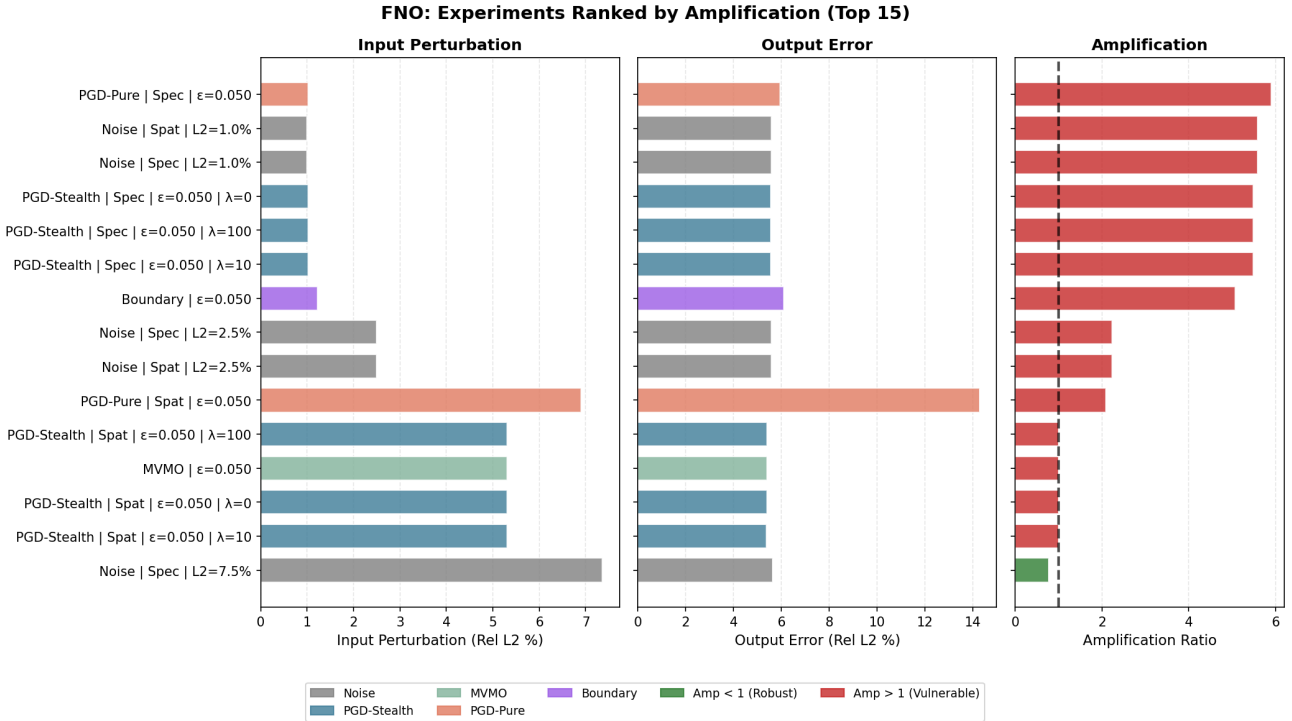


Figure 6. Output error versus absolute input perturbation FNO.

Table 7 below shows the raw data for all the experiments we performed, in terms of the relative input and output perturbations, across all the models.



Table 7. Complete Results: Darcy Flow Adversarial Attacks

| Method                 | Domain   | $\epsilon$ | CNO  |      |      | FNO  |      |       | FFNO |      |       |
|------------------------|----------|------------|------|------|------|------|------|-------|------|------|-------|
|                        |          |            | In   | Out  | Amp  | In   | Out  | Amp   | In   | Out  | Amp   |
| Clean                  | -        | -          | 0    | 2.0  | 1.0  | 0    | 3.5  | 1.0   | 0    | 2.9  | 1.0   |
| <i>Random Noise</i>    |          |            |      |      |      |      |      |       |      |      |       |
| Noise                  | Spatial  | 0.01       | 1.0  | 2.4  | 2.4  | 1.0  | 5.6  | 5.6   | 1.0  | 3.9  | 3.9   |
| Noise                  | Spatial  | 0.025      | 2.5  | 2.5  | 1.0  | 2.5  | 5.6  | 2.2   | 2.5  | 3.9  | 1.5   |
| Noise                  | Spatial  | 0.05       | 5.0  | 2.5  | 0.5  | 5.0  | 5.6  | 1.1   | 5.0  | 3.9  | 0.8   |
| Noise                  | Spatial  | 0.075      | 7.5  | 2.6  | 0.3  | 7.5  | 5.6  | 0.8   | 7.5  | 3.9  | 0.5   |
| Noise                  | Spatial  | 0.1        | 10.0 | 2.8  | 0.3  | 10.0 | 5.8  | 0.6   | 10.0 | 4.1  | 0.4   |
| Noise                  | Spectral | 0.01       | 1.0  | 2.4  | 2.4  | 1.0  | 5.6  | 5.6   | 1.0  | 3.9  | 3.9   |
| Noise                  | Spectral | 0.025      | 2.5  | 2.5  | 1.0  | 2.5  | 5.6  | 2.2   | 2.5  | 3.9  | 1.6   |
| Noise                  | Spectral | 0.05       | 5.0  | 2.5  | 0.5  | 5.0  | 5.6  | 1.1   | 5.0  | 3.9  | 0.8   |
| Noise                  | Spectral | 0.075      | 7.4  | 2.5  | 0.3  | 7.4  | 5.6  | 0.8   | 7.4  | 3.9  | 0.5   |
| Noise                  | Spectral | 0.1        | 9.6  | 2.7  | 0.3  | 9.6  | 5.7  | 0.6   | 9.6  | 3.9  | 0.4   |
| <i>PGD-Pure</i>        |          |            |      |      |      |      |      |       |      |      |       |
| PGD                    | Spatial  | 0.001      | 0.1  | 2.6  | 18.9 | 0.1  | 5.7  | 41.2  | 0.1  | 4.0  | 28.5  |
| PGD                    | Spatial  | 0.005      | 0.7  | 3.6  | 5.1  | 0.7  | 6.3  | 9.1   | 0.7  | 4.4  | 6.3   |
| PGD                    | Spatial  | 0.01       | 1.4  | 4.9  | 3.6  | 1.4  | 7.1  | 5.1   | 1.4  | 5.0  | 3.6   |
| PGD                    | Spatial  | 0.05       | 6.8  | 19.4 | 2.8  | 6.9  | 14.3 | 2.1   | 6.9  | 11.6 | 1.7   |
| PGD                    | Spatial  | 0.1        | 13.6 | 51.4 | 3.8  | 13.7 | 24.0 | 1.8   | 13.8 | 21.1 | 1.5   |
| PGD                    | Spectral | 0.001      | 1.0  | 3.0  | 2.9  | 1.0  | 5.9  | 5.8   | 1.0  | 4.1  | 4.0   |
| PGD                    | Spectral | 0.005      | 1.0  | 3.0  | 2.9  | 1.0  | 5.9  | 5.8   | 1.0  | 4.1  | 4.0   |
| PGD                    | Spectral | 0.01       | 1.0  | 3.0  | 2.9  | 1.0  | 5.9  | 5.8   | 1.0  | 4.1  | 4.0   |
| PGD                    | Spectral | 0.05       | 1.0  | 3.0  | 2.9  | 1.0  | 5.9  | 5.8   | 1.0  | 4.1  | 4.0   |
| PGD                    | Spectral | 0.1        | 1.0  | 3.0  | 2.9  | 1.0  | 5.9  | 5.8   | 1.0  | 4.1  | 4.0   |
| <i>PGD-Stealth</i>     |          |            |      |      |      |      |      |       |      |      |       |
| Stealth                | Spatial  | 0.001      | 0.1  | 2.4  | 17.4 | 0.1  | 5.6  | 40.1  | 0.1  | 3.9  | 27.8  |
| Stealth                | Spatial  | 0.005      | 0.6  | 2.4  | 3.8  | 0.6  | 5.5  | 8.8   | 0.6  | 3.9  | 6.5   |
| Stealth                | Spatial  | 0.01       | 1.3  | 2.5  | 2.0  | 1.3  | 5.5  | 4.4   | 1.2  | 3.8  | 3.2   |
| Stealth                | Spatial  | 0.05       | 4.7  | 4.7  | 1.0  | 5.3  | 5.4  | 1.0   | 4.2  | 3.9  | 0.9   |
| Stealth                | Spatial  | 0.1        | 6.7  | 5.3  | 0.8  | 8.8  | 5.6  | 0.6   | 6.5  | 3.8  | 0.6   |
| Stealth                | Spectral | 0.001      | 1.0  | 2.4  | 2.3  | 1.0  | 5.6  | 5.4   | 1.0  | 3.9  | 3.8   |
| Stealth                | Spectral | 0.005      | 1.0  | 2.4  | 2.3  | 1.0  | 5.6  | 5.4   | 1.0  | 3.9  | 3.8   |
| Stealth                | Spectral | 0.01       | 1.0  | 2.4  | 2.3  | 1.0  | 5.6  | 5.4   | 1.0  | 3.9  | 3.8   |
| Stealth                | Spectral | 0.05       | 1.0  | 2.4  | 2.3  | 1.0  | 5.6  | 5.4   | 1.0  | 3.9  | 3.8   |
| Stealth                | Spectral | 0.1        | 1.0  | 2.4  | 2.3  | 1.0  | 5.6  | 5.4   | 1.0  | 3.9  | 3.8   |
| <i>Boundary Attack</i> |          |            |      |      |      |      |      |       |      |      |       |
| Boundary               | -        | 0.001      | 0.0  | 2.4  | 99.8 | 0.0  | 5.6  | 228.5 | 0.0  | 3.9  | 158.6 |
| Boundary               | -        | 0.005      | 0.1  | 2.5  | 20.2 | 0.1  | 5.6  | 46.1  | 0.1  | 3.9  | 32.0  |
| Boundary               | -        | 0.01       | 0.2  | 2.5  | 10.2 | 0.2  | 5.7  | 23.2  | 0.2  | 4.0  | 16.2  |
| Boundary               | -        | 0.05       | 1.2  | 2.7  | 2.2  | 1.2  | 6.1  | 5.0   | 1.2  | 4.5  | 3.6   |
| Boundary               | -        | 0.1        | 2.3  | 3.0  | 1.3  | 2.4  | 6.7  | 2.8   | 2.5  | 5.2  | 2.1   |
| <i>MVMO Attack</i>     |          |            |      |      |      |      |      |       |      |      |       |
| MVMO                   | -        | 0.001      | 0.1  | 2.4  | 17.4 | 0.1  | 5.6  | 40.1  | 0.1  | 3.9  | 27.8  |
| MVMO                   | -        | 0.005      | 0.6  | 2.4  | 3.8  | 0.6  | 5.5  | 8.8   | 0.6  | 3.9  | 6.5   |
| MVMO                   | -        | 0.01       | 1.3  | 2.5  | 2.0  | 1.3  | 5.5  | 4.4   | 1.2  | 3.8  | 3.2   |
| MVMO                   | -        | 0.05       | 4.7  | 4.8  | 1.0  | 5.3  | 5.4  | 1.0   | 4.3  | 3.9  | 0.9   |
| MVMO                   | -        | 0.1        | 6.7  | 5.4  | 0.8  | 8.8  | 5.6  | 0.6   | 6.5  | 3.8  | 0.6   |

Figure 6 below shows a comparison of the top attacks by efficiency (amplification factor), based on the relative values of the  $L_2$  perturbation magnitudes. Note that the values in the report are averaged across the attack types, while here, the variance of the efficacy within a single attack type can be more obvious.

**Navier Stokes.** We present the same data attack rankings against FNO as well as a detailed table in Figure 7 and Table 8.

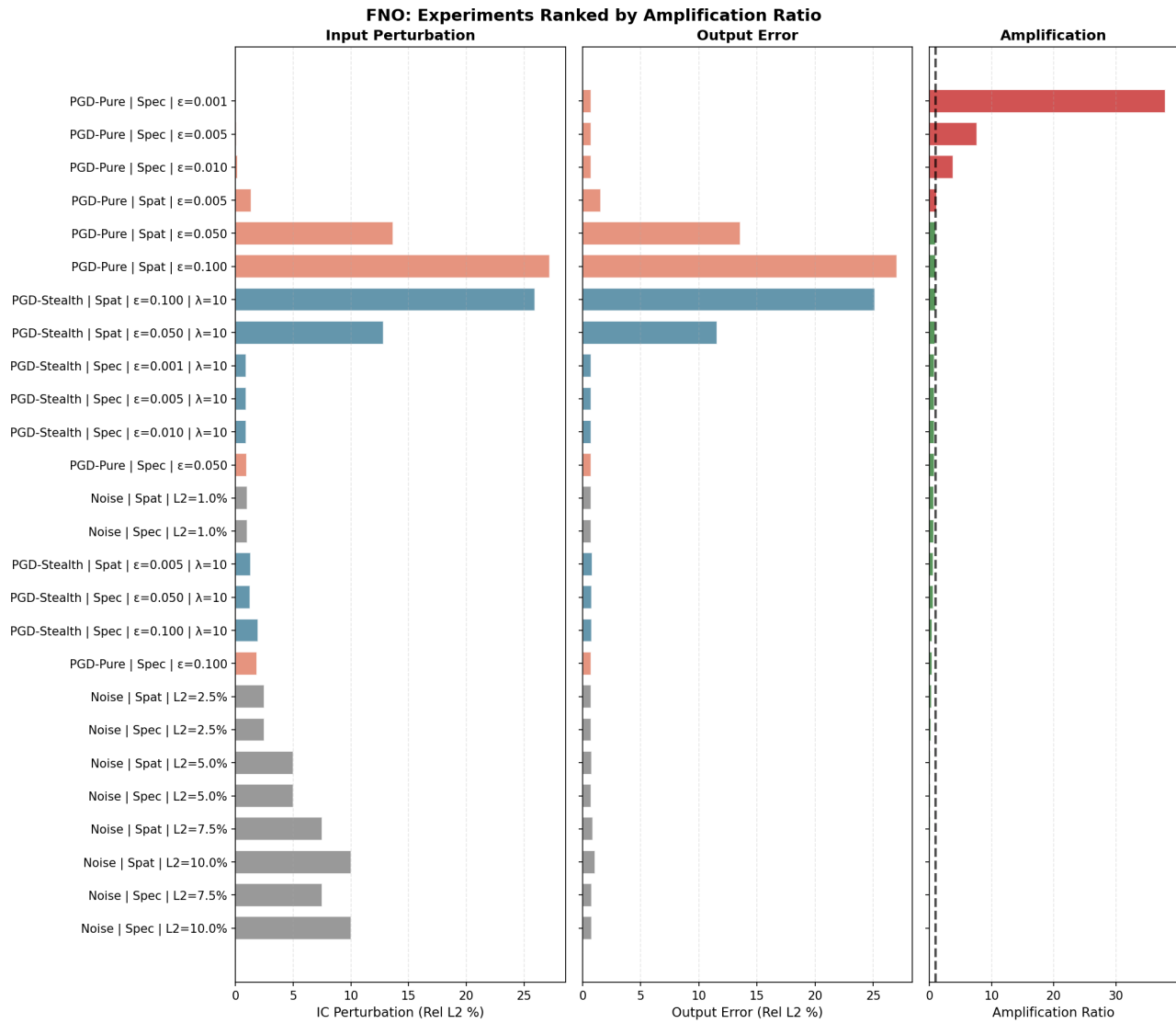


Figure 7. Output error versus absolute input perturbation FNO.

Table 8. Complete Results: Navier-Stokes Adversarial Attacks

| Method                                         | Domain   | $\epsilon$ | CNO  |      |      | FNO  |      |      | FFNO |      |      |
|------------------------------------------------|----------|------------|------|------|------|------|------|------|------|------|------|
|                                                |          |            | In   | Out  | Amp  | In   | Out  | Amp  | In   | Out  | Amp  |
| Clean                                          | -        | -          | 0    | 3.0  | 1.0  | 0    | 0.7  | 1.0  | 0    | 0.6  | 1.0  |
| <i>Random Noise</i>                            |          |            |      |      |      |      |      |      |      |      |      |
| Noise                                          | Spatial  | 0.01       | 1.0  | 2.6  | 2.6  | 1.0  | 0.7  | 0.7  | 1.0  | 0.5  | 0.5  |
| Noise                                          | Spatial  | 0.025      | 2.5  | 2.6  | 1.1  | 2.5  | 0.7  | 0.3  | 2.5  | 0.6  | 0.2  |
| Noise                                          | Spatial  | 0.05       | 5.0  | 2.7  | 0.5  | 5.0  | 0.8  | 0.2  | 5.0  | 0.7  | 0.1  |
| Noise                                          | Spatial  | 0.075      | 7.5  | 2.7  | 0.4  | 7.5  | 0.9  | 0.1  | 7.5  | 0.8  | 0.1  |
| Noise                                          | Spatial  | 0.1        | 10.0 | 3.0  | 0.3  | 10.0 | 1.1  | 0.1  | 10.0 | 0.9  | 0.1  |
| Noise                                          | Spectral | 0.01       | 1.0  | 2.6  | 2.6  | 1.0  | 0.7  | 0.7  | 1.0  | 0.5  | 0.5  |
| Noise                                          | Spectral | 0.025      | 2.5  | 2.6  | 1.1  | 2.5  | 0.7  | 0.3  | 2.5  | 0.5  | 0.2  |
| Noise                                          | Spectral | 0.05       | 5.0  | 2.6  | 0.5  | 5.0  | 0.7  | 0.1  | 5.0  | 0.6  | 0.1  |
| Noise                                          | Spectral | 0.075      | 7.5  | 2.7  | 0.4  | 7.5  | 0.8  | 0.1  | 7.5  | 0.6  | 0.1  |
| Noise                                          | Spectral | 0.1        | 10.0 | 2.9  | 0.3  | 10.0 | 0.8  | 0.1  | 10.0 | 0.7  | 0.1  |
| <i>PGD-Pure</i>                                |          |            |      |      |      |      |      |      |      |      |      |
| PGD                                            | Spatial  | 0.001      | 0.3  | 2.7  | 9.8  | 0.3  | 0.8  | 2.9  | 0.3  | 0.6  | 2.1  |
| PGD                                            | Spatial  | 0.005      | 1.4  | 3.5  | 2.5  | 1.4  | 1.6  | 1.2  | 1.4  | 1.4  | 1.1  |
| PGD                                            | Spatial  | 0.01       | 2.7  | 4.7  | 1.7  | 2.7  | 2.9  | 1.0  | 2.7  | 2.8  | 1.0  |
| PGD                                            | Spatial  | 0.05       | 13.6 | 17.9 | 1.3  | 13.6 | 13.6 | 1.0  | 13.6 | 13.7 | 1.0  |
| PGD                                            | Spatial  | 0.1        | -    | -    | -    | 27.3 | 27.0 | 1.0  | 27.2 | 26.7 | 1.0  |
| PGD                                            | Spectral | 0.001      | 0.1  | 2.6  | 41.6 | 0.0  | 0.7  | 37.9 | 0.0  | 0.5  | 27.8 |
| PGD                                            | Spectral | 0.005      | 0.1  | 2.6  | 18.1 | 0.1  | 0.7  | 7.7  | 0.1  | 0.5  | 5.6  |
| PGD                                            | Spectral | 0.01       | 0.3  | 2.5  | 9.8  | 0.2  | 0.7  | 3.8  | 0.2  | 0.5  | 2.8  |
| PGD                                            | Spectral | 0.05       | 1.0  | 2.6  | 2.5  | 0.9  | 0.7  | 0.8  | 0.9  | 0.5  | 0.6  |
| PGD                                            | Spectral | 0.1        | 2.0  | 2.9  | 1.5  | 1.9  | 0.7  | 0.4  | 1.9  | 0.6  | 0.3  |
| <i>PGD-Stealth (<math>\lambda = 10</math>)</i> |          |            |      |      |      |      |      |      |      |      |      |
| Stealth                                        | Spatial  | 0.001      | 0.3  | 2.5  | 9.1  | 0.3  | 0.7  | 2.7  | 0.3  | 0.5  | 2.0  |
| Stealth                                        | Spatial  | 0.005      | 1.4  | 2.3  | 1.7  | 1.3  | 0.8  | 0.6  | 1.2  | 0.7  | 0.6  |
| Stealth                                        | Spatial  | 0.01       | 2.7  | 2.7  | 1.0  | 2.6  | 1.3  | 0.5  | 2.4  | 1.3  | 0.5  |
| Stealth                                        | Spatial  | 0.05       | 11.2 | 7.9  | 0.7  | 12.8 | 11.6 | 0.9  | 12.2 | 10.8 | 0.9  |
| Stealth                                        | Spatial  | 0.1        | 19.6 | 13.5 | 0.7  | 25.9 | 25.1 | 1.0  | 24.6 | 23.5 | 1.0  |
| Stealth                                        | Spectral | 0.001      | 1.2  | 2.2  | 1.9  | 0.9  | 0.8  | 0.8  | 0.9  | 0.6  | 0.6  |
| Stealth                                        | Spectral | 0.005      | 1.2  | 2.2  | 1.9  | 0.9  | 0.8  | 0.8  | 0.9  | 0.6  | 0.6  |
| Stealth                                        | Spectral | 0.01       | 1.2  | 2.2  | 1.9  | 0.9  | 0.8  | 0.8  | 0.9  | 0.6  | 0.6  |
| Stealth                                        | Spectral | 0.05       | 1.5  | 2.2  | 1.5  | 1.3  | 0.8  | 0.6  | 1.2  | 0.6  | 0.5  |
| Stealth                                        | Spectral | 0.1        | 2.1  | 2.2  | 1.0  | 1.9  | 0.8  | 0.4  | 2.0  | 0.6  | 0.3  |
| <i>Sequential</i>                              |          |            |      |      |      |      |      |      |      |      |      |
| Seq.                                           | Spatial  | 0.001      | 0.0  | 4.0  | 0.0  | 0.0  | 3.2  | 0.0  | 0.0  | 3.0  | 0.0  |
| Seq.                                           | Spatial  | 0.005      | 0.0  | 15.1 | 0.0  | 0.0  | 14.5 | 0.0  | 0.0  | 14.3 | 0.0  |
| Seq.                                           | Spatial  | 0.01       | 0.0  | 28.0 | 0.0  | 0.0  | 30.3 | 0.0  | 0.0  | 29.6 | 0.0  |
| Seq.                                           | Spatial  | 0.05       | -    | -    | -    | -    | -    | -    | -    | -    | -    |
| Seq.                                           | Spatial  | 0.1        | -    | -    | -    | -    | -    | -    | -    | -    | -    |
| Seq.                                           | Spectral | 0.001      | 0.0  | 2.9  | 0.0  | 0.0  | 1.7  | 0.0  | 0.0  | 1.2  | 0.0  |
| Seq.                                           | Spectral | 0.005      | 0.0  | 2.9  | 0.0  | 0.0  | 1.7  | 0.0  | 0.0  | 1.2  | 0.0  |
| Seq.                                           | Spectral | 0.01       | 0.0  | 2.9  | 0.0  | 0.0  | 1.7  | 0.0  | 0.0  | 1.2  | 0.0  |
| Seq.                                           | Spectral | 0.05       | 0.0  | 2.8  | 0.0  | 0.0  | 1.6  | 0.0  | 0.0  | 1.1  | 0.0  |
| Seq.                                           | Spectral | 0.1        | 0.0  | 2.6  | 0.0  | 0.0  | 1.7  | 0.0  | 0.0  | 1.2  | 0.0  |

## D. Appendix: Individual Darcy Flow perturbation samples

In this section, we will provide an example of a clean Darcy Input (Figure 8 and a boundary condition perturbation (Figure 9).

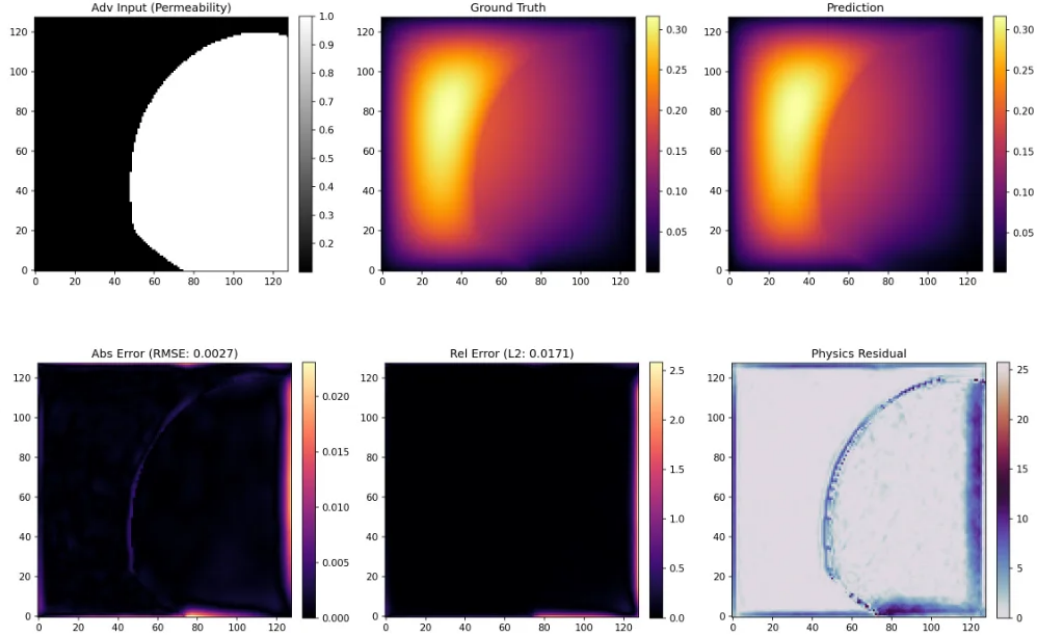


Figure 8. Sample clean input and prediction for the Darcy Flow.

In the clean sample, the error is distributed uniformly across the entire image, with small perturbations on the boundaries, while in the perturbed sample, the relative error on the left boundary is significantly higher.

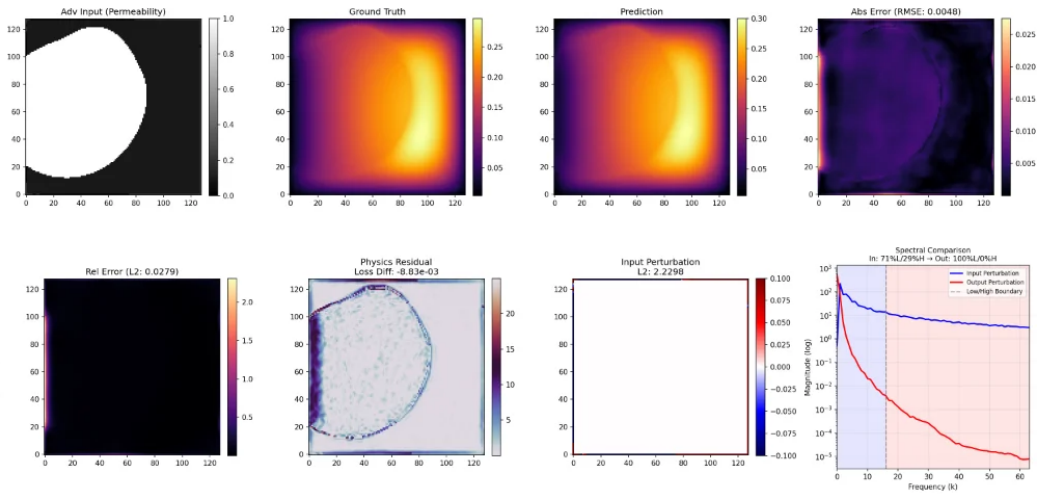


Figure 9. Sample perturbed input and predictions for an FNO model on a boundary attack.

## E. Appendix: Individual Navier Stokes perturbation samples

We present an evolution of an spatial input perturbation and an example of how the Navier-Stokes FNO model manages to recover over time, as the FNO successfully simulates the physical laws governing the simulated fluid. The physical residual and absolute error is magnified at  $T = 0$ , and slowly dissipates towards  $T = 50$ .

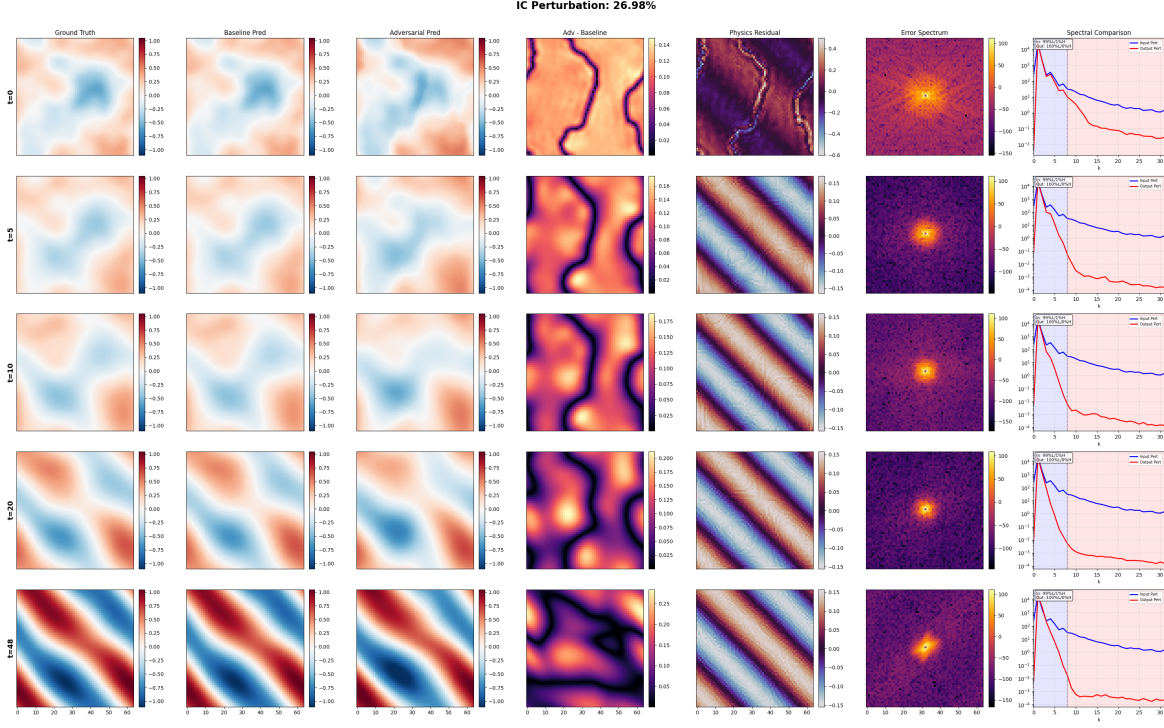


Figure 10. Snapshots of the perturbation impact at different timesteps for Navier Stokes.